

$\therefore$ MCQs :-

$$\frac{0.0000728}{7.28 \times 10^{-5}}$$

1. B
2. B
3. A
4. C
5. C
6. A
7. A
8. A
9. A
10. A
11. B
12. B

Q: 2(i)(a)

$$\text{Yards} = 3.2$$

$$\text{Price} = 139.2$$

Cost : ?

$$\frac{139.2}{3.2}$$

$$= 43.5$$

Q: 2(ii)

$$\text{Balloons} = 42$$

$$\text{Cost} = 1.85 \text{ each}$$

$$\text{total amount} = 42 \times 1.85$$

$$77.7 \text{ RS.}$$

Q.2(ii)(or)

~~distance~~ distance: ~~283.4~~ 283.4 Km

litres: - 16.2

averaged speed: - 17.5

Let's make (17.5) reasonable:-

$$\frac{283.4}{16.2}$$

$$= 17.49 \approx 17.5$$

It's reasonable

Q.2(iv)(or)

(a) 3.5636

Characteristic: - 3

mantissa: - 3656 + 5

3661

~~3661~~ 3661.0

antilog 3.5636 $\approx$ 3661

(b) 2.9281

Characteristic: - 2

mantissa: - 8472 + 2

8474

847.4

Q2(v)

seconds: 5 (once)

1 million: $5 \times 1,000,000$
 $5,000,000$ s

days (seconds)

$$\frac{5,000,000}{86400}$$

$$= 1388.8$$

$$\text{In 24hr} \therefore \frac{1388.8}{24}$$

$$57.86$$

Round off:-

~~57~~ 58 days.

Q: 2(vi)(or)

$$2n^2 - 5n - 3$$

$$2n^2 - 6n + 1n - 3$$

$$2n(n-3) + 1(n-3)$$

$$(2n+1)(n-3)$$

$$2(vii)(or) -$$

$$2(vii)(or) -$$

$$2x^2 + 5x - 3$$

$$2x^2 + 6x - 1x - 3$$

$$2x(x+3) - 1(x+3)$$

$$(2x-1)(x+3)$$

$$x^2 + 2x - 3$$

$$x^2 - x + 3x - 3$$

$$x(x-1) + 3(x-1)$$

$$(x+3)(x-1)$$

$$LCF :: (x+3)$$

(3 - 3)

Q: 2 (viii) (00) :-

$$3 \mid 2 - 2 \mid - 4 = -2$$

$$3 \mid 2 - 2 \mid = -2 + 4$$

$$3 \mid 2 - 2 \mid = 2$$

$$\mid 2 - 2 \mid = \frac{2}{3}$$

$$2 - 2 = \frac{2}{3}, \quad 2 - 2 = -\frac{2}{3}$$

$$2 = \frac{2}{3} + \frac{2 \times 3}{1 \times 3}$$

$$2 = -\frac{2}{3} + \frac{2 \times 3}{1 \times 3}$$

$$2 = \frac{2 + 6}{3}$$

$$2 = \frac{-2 + 6}{3}$$

$$2 = \frac{8}{3}$$

$$2 = \frac{+4}{3}$$

$$\left\{ \frac{8}{3}, \frac{+4}{3} \right\}$$

Q: 2 (in)
A: 2 (in)

$$3x + 2 < 1 - 2 \quad | -x$$

$$4x < -20$$

$$x < -5$$

$$x < 5$$

$$\text{or } 3x + 8 > 3 - 2x$$

$$3x + 2x > 3 - 8$$

$$5x > -5$$

$$x > -1$$

$$\{x | x \in \mathbb{R}, x < 5 \text{ or } x > -1\}$$

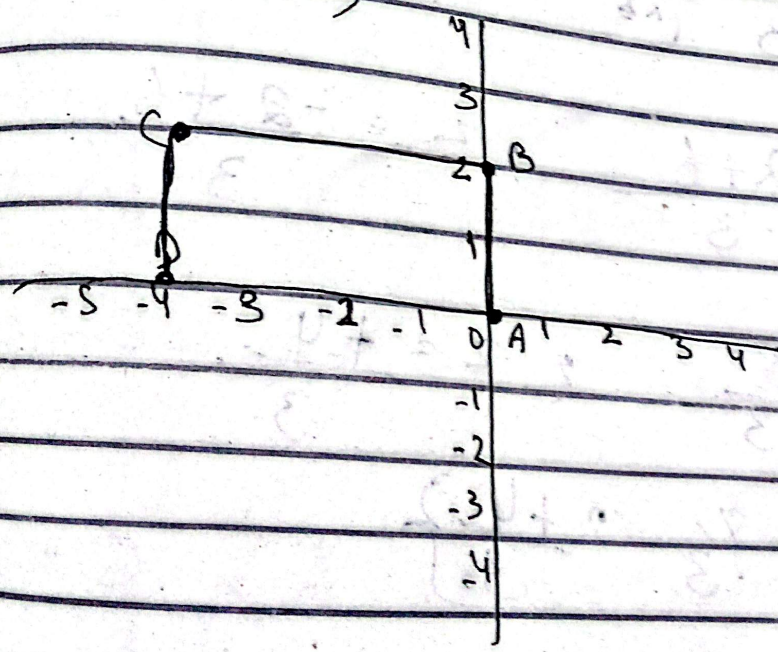
Q: 2 (n) (or)
A: 2 (n) (or)

$$B = (0, 2)$$

$$D = (-4, 0)$$

$$A = (0, 0)$$

$$C = (-4, 2)$$



Q: 2 (ni)

A(3, 4)

B(-1, 7)

C(x, y)

B is the midpoint of AC ∴

A(3, 4) C(x, y)

$$\frac{3+x}{2}, \frac{4+y}{2} = (-1, 7)$$

$$\frac{3+x}{2} = -1, \frac{4+y}{2} = 7$$

$$3+x = -2, 4+y = 14$$

$$3+x = -2, 4+y = 14$$

$$x = -2 - 3, y = -4 + 14$$

$$x = -5, y = 10$$

$$C = (-5, 10)$$

$$Q3(i)$$

$$A: 3(i)$$

$$\frac{6.3}{20}$$

$$\frac{9 \times 7}{4 \times 5}$$

$$= \frac{3^2 \times 7}{(2)^2 \times 5}$$

$$\log 3^2 + \log 7 - \log 2^2 - \log 5$$

$$2 \log 3 + \log 7 - 2 \log 2 - \log 5$$

$$2(0.4771) + 0.8451 - 2(0.3010) - 0.6990$$

$$0.9542 + 0.8451 - 0.602 - 0.6990$$

$$= 0.4983$$

$$b) \log 6$$

$$\log (2 \times 3)$$

$$\log 2 + \log 3$$

$$0.3010 + 0.4771$$

$$0.7781$$

$$\log 6)^{45}$$

$$45 \times 0.7781 = 35.0145$$

$$\text{Characteristic} = 35 + 1$$

$$36$$

Q: 3(ii) (or)
A: 3(ii) (or)

$$\text{Area} = 25x^2 - 30x + 9$$

$$= (5x)^2 - 2(5x)(3) + (3)^2$$

$$= (5x - 3)^2$$

(a) Side length = $(5x - 3)m$

(b) Boundary (perimeter) = $4(5x - 3)$
 $(20x - 12)m$

(c) Cost of polishing :-

$$RS(1875x^2 - 2250x + 675)$$

(d) Cost of edging 2 sides =

$$28 \times 2(5x - 3) = 280x - 168$$

$$RS(280x - 168)$$

Q: 3(iv)

A: 3(iii)

$$-4 \leq 3x \leq 22$$

{ subtract 5
from all parts }

$$-4 \leq 3x \leq 17$$

$$\frac{-4}{3} \leq x \leq \frac{17}{3}$$

$$\{ x \mid x \in \mathbb{R}, -\frac{4}{3} \leq x \leq \frac{17}{3} \}$$

Number line :-

Solid dots at $-\frac{4}{3}$ and $\frac{17}{3}$,
shaded segments b/w them.

$$2) |5y| = 9$$

$$5y = 9 \Rightarrow 5y = -9$$

$$y = \frac{9}{5} \Rightarrow y = -\frac{9}{5}$$

$$Q: 3(iv) \text{ (or)}$$

$$A: 3(iv) \text{ (or)}$$

$$\begin{aligned} dAB &= \sqrt{(7-5)^2 + (5-8)^2} \\ &= \sqrt{(2)^2 + (-3)^2} \\ &= \sqrt{4+9} \\ &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} dBC &= \sqrt{(5-7)^2 + (2-5)^2} \\ &= \sqrt{(-2)^2 + (-3)^2} \\ &= \sqrt{4+9} \\ &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} dCD &= \sqrt{(3-5)^2 + (5-2)^2} \\ &= \sqrt{(-2)^2 + (+3)^2} \\ &= \sqrt{4+9} \\ &= \sqrt{13} \end{aligned}$$

$$\begin{aligned} dDA &= \sqrt{(5-3)^2 + (8-5)^2} \\ &= \sqrt{(2)^2 + (3)^2} \\ &= \sqrt{4+9} \\ &= \sqrt{13} \end{aligned}$$

4 sides (four) are equal
so ABCD is rhombus.

Diagonals:

$$\begin{aligned} d_{AC} &:= \sqrt{(5-5)^2 + (2-8)^2} \\ &= \sqrt{(0)^2 + (-6)^2} \\ &= \sqrt{0 + 36} \\ &= \sqrt{36} \Rightarrow 6 \end{aligned}$$

$$\begin{aligned} d_{BD} &:= \sqrt{(3-7)^2 + (5-5)^2} \\ &= \sqrt{(-4)^2 + (0)^2} \\ &= \sqrt{16} \\ &= 4 \end{aligned}$$